

## Units of time



1 Convert the following into 24-hour time:

- |            |            |            |            |
|------------|------------|------------|------------|
| a 11:33 am | b 11:26 pm | c 12:56 am | d 10:30 am |
| e 9:15 pm  | f 8:40 am  | g 7:25 pm  | h 10:20 pm |

2 Convert the following into 12-hour time:

- |         |         |         |         |
|---------|---------|---------|---------|
| a 20:59 | b 23:10 | c 10:44 | d 07:12 |
| e 18:45 | f 02:00 | g 14:24 | h 00:30 |

3 Express the following times in 24-hour time:

- |                                  |                                       |
|----------------------------------|---------------------------------------|
| a Quarter to midnight.           | b Twenty past one in the afternoon.   |
| c Half past ten at night.        | d Quarter past eleven in the morning. |
| e Ten past eight in the morning. | f Ten to four in the afternoon.       |

4 A clock shows the time 19:39. Describe the time being shown in words.

5 Write the time shown on each of the following clocks in 24-hour time:

a



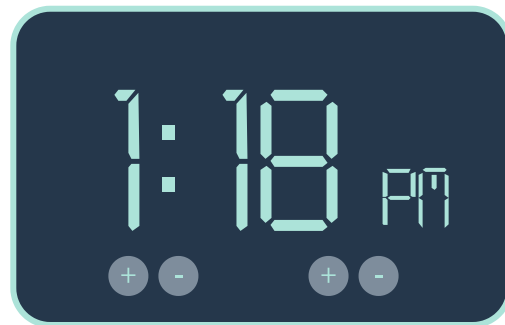
b



c



d



6 Convert the following to seconds:

- |                          |                           |              |               |
|--------------------------|---------------------------|--------------|---------------|
| a 10 minutes             | b 3.5 minutes             | c 16 hours   | d 9.6 minutes |
| e $1\frac{1}{4}$ minutes | f $12\frac{3}{4}$ minutes | g 11 minutes | h 24 hours    |

7 Convert the following to minutes:

- |               |                |               |               |
|---------------|----------------|---------------|---------------|
| a 660 seconds | b 1320 seconds | c 360 seconds | d 600 seconds |
| e 5 hours     | f 8.6 hours    | g 11.5 hours  | h 12 hours    |

8 Convert the following to hours:

- |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|
| a 360 minutes    | b 600 minutes    | c 780 minutes    | d 594 minutes    |
| e 36 000 seconds | f 28 800 seconds | g 11 160 seconds | h 25 920 seconds |

9 Complete the following statements:

- a 253 seconds is equal to  minutes and  seconds.
- b 285 minutes is equal to  hours and  minutes.
- c 4545 seconds is equal to  hour,  minutes, and  seconds.

10 Convert the following. Assume that there are 30 days in a month and 52 weeks in a year.

- |                             |                      |
|-----------------------------|----------------------|
| a 23 days to hours.         | b 11 days to hours.  |
| c A non-leap year to hours. | d 2 weeks to days.   |
| e 312 hours to days.        | f 4 months to days.  |
| g 5 fortnights to days.     | h 70 days to weeks.  |
| i 84 days to weeks.         | j 3 years to weeks.  |
| k 2 years to fortnights.    | l 2 years to months. |

- 11 Find the number of hours in 5 months. Assume that there are 30 days in a month.



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## Applications

- 12 A cake needs to cook for one and a half hours. How many minutes is this?
- 13 Han boils an egg for 360 seconds. How many minutes is this?
- 14 It takes around 300 minutes to drive from Sydney to Thredbo. How many hours is this?
- 15 The walk to the letterbox and back takes 1 minute and 48 seconds. How many seconds does the walk take in total?
- 16 Tamba ran a marathon in 2 hours and 42 minutes. How many minutes was this?
- 17 Nadia drove for 2 hours and 52 minutes. How many minutes was this?
- 18 An astronomer observes that a newly discovered planet rotates once every 67 hours, 25 minutes, 3 seconds. How often does the planet rotate in seconds?
- 19 Carl finished a marathon in 2 hours, 35 minutes and 19 seconds. How many seconds is that?
- 20 Dave just turned 9 years old. Assuming that there are 365 days in a year and ignoring leap years, find his age in:
- a Days                      b Hours                      c Minutes                      d Seconds
- 21 If an average person sleeps for 8 hours a night, how many weeks does an average person spend asleep in 65 years? Ignore leap years and round your answer to one decimal place.
- 22 If an average person showers for 6 minutes per day, how many hours do they spend in a shower in:
- a A non-leap year?                      b 55 years (ignoring leap years)?